Final project

Aditya Chander and
Joseph Tran

Hypothesis: background

- Autism Spectrum Disorders (ASD): challenges in social communication and creative behaviours

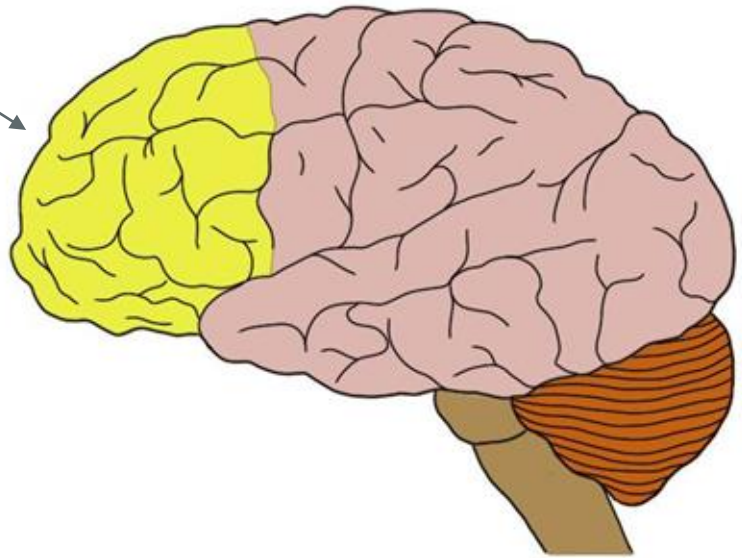
- Music therapy: joint improvisation

Hypothesis: desired consequences

- Inspire ASD individuals to expand creative behaviours through musical improvisation
- Secondary benefit: promote emotional recognition
- We hope that these effects will promote a better long-term quality of life
- Bridge to actual music therapy

Hypothesis: neural correlates

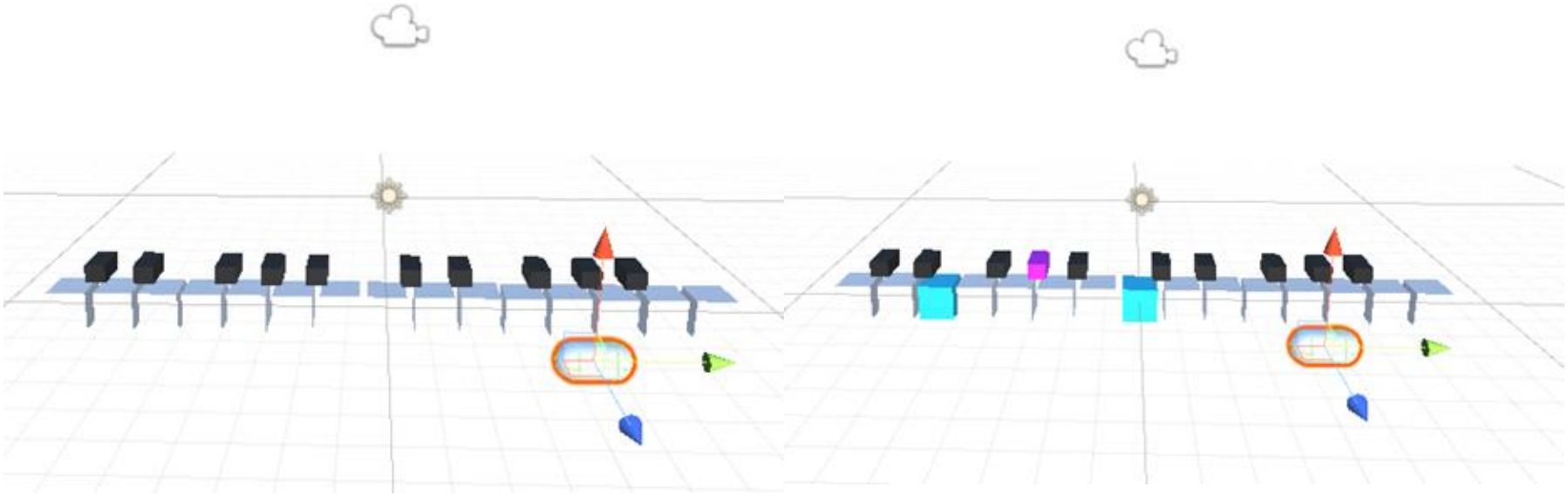
- Prefrontal cortex (PFC)
- Implicated in musical improvisation
- ASD sufferers exhibit PFC structural anomalies
- May expect changed activation patterns here



Objective of the game

- Produce novel melodies on a MIDI keyboard
- 12 bar blues harmonic backing
- Stages
  - Initially, produce a fixed melody to repeat
  - Then provide approximate contour of melody, several note options permissible but some are better than others. Rhythm fixed
  - Then remove contour specification but continue to provide a rhythm
  - Finally, remove the rhythm

Keyboard



Enjoyment and measuring improvement

- Fun mystery planet environment with piano as a controller to defeat enemies
  - If you don’t defeat enough enemies you lose a life and have to repeat the level

- Point scoring system: having a goal to reach is important
  - Fixed rhythm: 1 or 2 points for a correct note, depending on the time window in which the key was pressed, -1 for out of time note or wrong pitch
  - Improvisation: 1 or 2 points for a well-fitting note, -1 for a poorly fitting note, -2 if it also falls outside the time tolerance window
  - Streak bonus; incremental depending on length of streak

- Visual feedback: emotionally valenced faces

Expected results

- Improvement in gameplay after several rounds, ideally corresponds to more creative behaviours

- Perhaps measure before and after using S-A creativity test (measuring fluency, flexibility, originality, elaboration)

- Facial recognition (secondary goal)